

- 6 (a) (i) Random means cannot predict when the decay will occur, or which nuclide will decay. **B1**

OR

Constant probability of decay per unit time **B1**

- (ii) Spontaneous means occurs without any external stimuli such as changes in temperature or pressure. **B1**

MC:	For (a)(i) and (a)(ii), these are very commonly tested definitions, Hence, little variance in answers was accepted. Students are advised to be familiar with these definitions by hard. Generally not very well done. Most students had an idea of what the random and spontaneous mean, but many responses tended to have missing parts/keywords or students attempted to phrase it in their own way.
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- (b) (i) Gamma radiation has a greater penetrating power and will penetrate through a thin sheet of aluminium foil with little loss in intensity. **B1**

Any differences in the intensity of gamma radiation detected by the detector due to variations in the foil thickness would be too small to detect easily. **B1**

MC:	Many candidates were at least able to hint at the penetrating power being greater for gamma radiation. However, usage of keywords such as “penetrating power” was generally very poorly utilized. Most candidates ended up listing examples to illustrate this point. Most students also fell short in re-contextualisation of their answers to answer the question. Most just stated that the penetrating power is greater without making the link or explaining how it would affect the operation of the setup in the question. Hence, most candidates could only score partial credit at best. Common misconceptions: • All gamma radiation will pass through the foil regardless of thickness (false – there is attenuation albeit a very small/negligible amount, which does alter the count rate received by the detector) • The machine will recognize an increase in thickness if the count rate of beta particles is zero. (not necessarily true as the machine can operate by ensuring a fixed count rate, thickening of the foil can cause the count rate to decrease but not necessarily reach zero)
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(ii)

$$\begin{aligned}
 \lambda &= \frac{\ln 2}{t_{1/2}} \\
 &= \frac{\ln 2}{14 \times 24 \times 60 \times 60} \\
 &= 5.7303 \times 10^{-7} \\
 &= 5.7 \times 10^{-7} \text{ Bq}
 \end{aligned}$$

C1

A1

(iii)

$$\begin{aligned}
 A &= A_0 e^{-\lambda t} \\
 \frac{A}{A_0} &= e^{-\lambda t} \\
 &= e^{-\left(\frac{\ln 2}{14 \times 24}\right) 8} \\
 &= \left(\frac{1}{2}\right)^{\frac{8}{14 \times 24}} \\
 &= 0.98363 \\
 &= 0.98 \text{ (to 2sf)}
 \end{aligned}$$

C1

A1

MC:	(ii) and (iii) were generally well answered. A vast majority who attempted were able to score full credit for these two parts.
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(iv) When the detector detects a lower count rate, the set up interprets the decrease in count rate as a thickening of the foil. To maintain a constant foil thickness at the original calibrated value, the separation of the rollers decreases.

B1

Hence, the foil thickness decreases.

B1